You said:

: Generate the `readStream / withWatermark / groupBy(window(...))` skeleton with an explicit schema; audit it for schema inference, missing watermarks, unbounded state, and non-idempotent sinks. // how do I do this ?

ChatGPT said:

You said:

the prof said for streaming we can use a tuple with timtestamps

ChatGPT said:

You said:

why should we use transactiondt, this is all fake anyway, so let's create a convinient time

ChatGPT said:

You said:

here are further instructions: Replay your records as rate-controlled micro-batches into a file-source landing directory,// can you take me step by step, I really want to learn, and not get into ai slop.

ChatGPT said:

You said:

```
+-------------+--------------+---------+
|TransactionID|TransactionAmt|ProductCD|
+-------------+--------------+---------+
|      2987000|          68.5|        W|
|      2987001|          29.0|        W|
|      2987002|          59.0|        W|
|      2987003|          50.0|        W|
```

```
|     2987004|         50.0|       H|
+------------+-------------+--------+
only showing top 5 rows//path
dbfs:/Volumes/workspace/default/ieee_fraud_bronze/mp5_landing/batch_000000/
dbfs:/Volumes/workspace/default/ieee_fraud_bronze/mp5_landing/batch_000001/
dbfs:/Volumes/workspace/default/ieee_fraud_bronze/mp5_landing/batch_000002/
dbfs:/Volumes/workspace/default/ieee_fraud_bronze/mp5_landing/batch_000003/
dbfs:/Volumes/workspace/default/ieee_fraud_bronze/mp5_landing/batch_000004/
dbfs:/Volumes/workspace/default/ieee_fraud_bronze/mp5_landing/batch_000005/
dbfs:/Volumes/workspace/default/ieee_fraud_bronze/mp5_landing/batch_000006/
dbfs:/Volumes/workspace/default/ieee_fraud_bronze/mp5_landing/batch_000007/
dbfs:/Volumes/workspace/default/ieee_fraud_bronze/mp5_landing/batch_000008/
dbfs:/Volumes/workspace/default/ieee_fraud_bronze/mp5_landing/batch_000009/
dbfs:/Volumes/workspace/default/ieee_fraud_bronze/mp5_landing/batch_000010/
dbfs:/Volumes/workspace/default/ieee_fraud_bronze/mp5_landing/batch_000011/
```

ChatGPT said:

You said:

let's just delete all the old data, and start again

ChatGPT said:

You said:

empty list now

ChatGPT said:

You said:

Help me write these 10 transactions to batch_000000 as a Parquet file, then read it back and verify the rows. Explain each step so I learn.

ChatGPT said:

You said:

[**UNSUPPORTED_OPERATION**] errorifexists is not supported.

ChatGPT said:

You said:

Everythings working fine, can you explain how this fits into the assignment

ChatGPT said:

You said:

Help me write the rate-controlled replay loop that writes the next 10 transactions to a new Parquet batch every few seconds. Explain each step so I learn.

ChatGPT said:

You said:

Help me build the readStream consumer for my replayed Parquet batches, step by step, and explain the explicit schema.

ChatGPT said:

You said:

yes

ChatGPT said:

You said:

Help me write the streaming output sinks for both window aggregations, with separate checkpoint directories, and explain each step.

ChatGPT said:

You said:

[**INFINITE_STREAMING_TRIGGER_NOT_SUPPORTED**] Trigger type ProcessingTime is not supported for this cluster type.

ChatGPT said:

You said:

Query ID: bc5ab0af-9ba0-48de-93eb-7527191eeda0
Active: False
Status: {'message': 'Stopped', 'isDataAvailable': False, 'isTriggerActive': False}
Last progress: {
  "id": "bc5ab0af-9ba0-48de-93eb-7527191eeda0",
  "runId": "4675bdae-bcc8-4041-9e79-578ef7a9ebb9",
  "name": null,
  "timestamp": "2026-10-21T20:46:25.984Z",
  "batchId": 0,
  "batchDuration": 59,
  "durationMs": {
    "triggerExecution": 54,
    "latestOffset": 20
  },
  "eventTime": {},

```
  "stateOperators": [],
  "sources": [
    {
      "description":
"FileStreamSource[dbfs:/Volumes/workspace/default/ieee_fraud_bronz
e/mp5_landing]",
      "startOffset": null,
      "endOffset": null,
      "latestOffset": null,
      "numInputRows": 0,
      "inputRowsPerSecond": 0.0,
      "processedRowsPerSecond": 0.0,
      "metrics": {}
    }
  ],
  "sink": {
    "description":
"FileSink[/Volumes/workspace/default/ieee_fraud_bronze/mp5_output
s/tumbling_60s]",
    "numOutputRows": -1,
    "metrics": {}
  },
  "observedMetrics": {},
  "rtmMetrics": null
}
-------------------------------------------------
Query ID: cc40a977-3389-4805-9824-35ba9360bd0a
Active: False
Status: {'message': 'Stopped', 'isDataAvailable': False, 'isTriggerActive':
False}
Last progress: {
  "id": "cc40a977-3389-4805-9824-35ba9360bd0a",
  "runId": "97d80b92-e453-46ee-89f8-2c551726636d",
  "name": null,
  "timestamp": "2026-10-21T20:47:46.512Z",
  "batchId": 0,
  "batchDuration": 0,
  "durationMs": {
    "triggerExecution": 0,
    "latestOffset": 0
  },
  "eventTime": {},
```

```
  "stateOperators": [],
  "sources": [
    {
      "description":
"FileStreamSource[dbfs:/Volumes/workspace/default/ieee_fraud_bronz
e/mp5_landing]",
      "startOffset": null,
      "endOffset": null,
      "latestOffset": null,
      "numInputRows": 0,
      "inputRowsPerSecond": 0.0,
      "processedRowsPerSecond": 0.0,
      "metrics": {}
    }
  ],
  "sink": {
    "description":
"FileSink[/Volumes/workspace/default/ieee_fraud_bronze/mp5_output
s/sliding_5m_1m]",
    "numOutputRows": -1,
    "metrics": {}
  },
  "observedMetrics": {},
  "rtmMetrics": null
}
-------------------------------------------------
```


ChatGPT said:

You said:


```
path name  size  modificationTime
dbfs:/Volumes/workspace/default/ieee_fraud_bronze/mp5_landing/ba
tch_000000/  batch_000000/  0  1792615785877
dbfs:/Volumes/workspace/default/ieee_fraud_bronze/mp5_landing/ba
tch_000001/  batch_000001/  0  1792615785877
dbfs:/Volumes/workspace/default/ieee_fraud_bronze/mp5_landing/ba
tch_000002/  batch_000002/  0  1792615785877
dbfs:/Volumes/workspace/default/ieee_fraud_bronze/mp5_landing/ba
tch_000003/  batch_000003/  0  1792615785877/////TransactionID
TransactionAmt  isFraud  event_ts
2987000  68.5  0  2025-01-01T00:00:00.000+00:00
```

```
2987001  29    0  2025-01-01T00:00:00.000+00:00
2987002  59    0  2025-01-01T00:00:00.000+00:00
2987003  50    0  2025-01-01T00:00:00.000+00:00
2987004  50    0  2025-01-01T00:00:00.000+00:00
2987005  49    0  2025-01-01T00:00:00.000+00:00
2987006  159   0  2025-01-01T00:00:00.000+00:00
2987007  422.5   0  2025-01-01T00:00:00.000+00:00
2987008  15    0  2025-01-01T00:00:00.000+00:00
2987009  117   0  2025-01-01T00:00:00.000+00:00
```

ChatGPT said:

You said:

```
Total rows: 40
Distinct transactions: 40
Batch directories: 4//+-------------------+-------------------+-----------
---------+
|earliest_event     |latest_event       |distinct_event_times|
+-------------------+-------------------+--------------------+
|2025-01-01 00:00:00|2025-01-01 00:03:00|4                   |
+-------------------+-------------------+--------------------+
```

ChatGPT said:

You said:

```
Tumbling input rows: None
Sliding input rows: None
```

ChatGPT said:

You said:

```
umbling active: False
Tumbling status: {'message': 'Stopped', 'isDataAvailable': False,
'isTriggerActive': False}
Tumbling exception: None
Tumbling recent progress: [{
    "id": "7f6e9946-ce9d-4c8b-92e0-6d0bfeb4f962",
```

```
  "runId": "2eb7e66a-1fe4-41f5-b2f3-168346603048",
  "name": null,
  "timestamp": "2026-10-21T20:52:50.131Z",
  "batchId": 0,
  "batchDuration": 26130,
  "durationMs": {
    "triggerExecution": 26129,
    "queryPlanning": 1409,
    "collectSourceMetrics": 0,
    "getBatch": 477,
    "commitOffsets": 106,
    "addBatch": 23800,
    "latestOffset": 115,
    "walCommit": 321
  },
  "eventTime": {
```


ChatGPT said:

You said:


```
--- Tumbling ---
Batch ID: 1
Input rows: None
Sources: [{
  "description":
"FileStreamSource[dbfs:/Volumes/workspace/default/ieee_fraud_bronze/mp5_landing]",
  "startOffset": "{\"logOffset\":0}",
  "endOffset": "{\"logOffset\":0}",
  "latestOffset": null,
  "numInputRows": 0,
  "inputRowsPerSecond": 0.0,
  "processedRowsPerSecond": 0.0,
  "metrics": {}
}]
Sink: {
  "description":
"FileSink[/Volumes/workspace/default/ieee_fraud_bronze/mp5_outputs_retry1/tumbling_60s]",
  "numOutputRows": -1,
  "metrics": {}
```

```
}

--- Sliding ---
Batch ID: 1
Input rows: None
Sources: [{
  "description":
"FileStreamSource[dbfs:/Volumes/workspace/default/ieee_fraud_bronz
e/mp5_landing]",
  "startOffset": "{\"logOffset\":0}",
  "endOffset": "{\"logOffset\":0}",
  "latestOffset": null,
  "numInputRows": 0,
  "inputRowsPerSecond": 0.0,
  "processedRowsPerSecond": 0.0,
  "metrics": {}
}]
Sink: {
  "description":
"FileSink[/Volumes/workspace/default/ieee_fraud_bronze/mp5_output
s_retry1/sliding_5m_1m]",
  "numOutputRows": -1,
  "metrics": {}
}
```

ChatGPT said:

You said:

```
Top-level landing directory:  path  name  size  modificationTime
dbfs:/Volumes/workspace/default/ieee_fraud_bronze/mp5_landing/ba
tch_000000/  batch_000000/  0  1792616228433
dbfs:/Volumes/workspace/default/ieee_fraud_bronze/mp5_landing/ba
tch_000001/  batch_000001/  0  1792616228433
dbfs:/Volumes/workspace/default/ieee_fraud_bronze/mp5_landing/ba
tch_000002/  batch_000002/  0  1792616228433
dbfs:/Volumes/workspace/default/ieee_fraud_bronze/mp5_landing/ba
tch_000003/  batch_000003/  0  1792616228433///////path name
size  modificationTime
dbfs:/Volumes/workspace/default/ieee_fraud_bronze/mp5_landing/ba
tch_000000/  batch_000000/  0  1792616228433
dbfs:/Volumes/workspace/default/ieee_fraud_bronze/mp5_landing/ba
```

tch_000001/ batch_000001/ 0 1792616228433
dbfs:/Volumes/workspace/default/ieee_fraud_bronze/mp5_landing/ba
tch_000002/ batch_000002/ 0 1792616228433
dbfs:/Volumes/workspace/default/ieee_fraud_bronze/mp5_landing/ba
tch_000003/ batch_000003/ 0 1792616228433

ChatGPT said:

You said:

path name size modificationTime
dbfs:/Volumes/workspace/default/ieee_fraud_bronze/mp5_landing/ba
tch_000000/_SUCCESS _SUCCESS 0 1792613663000
dbfs:/Volumes/workspace/default/ieee_fraud_bronze/mp5_landing/ba
tch_000000/_committed_8299349501481159316
_committed_8299349501481159316 124 1792613663000
dbfs:/Volumes/workspace/default/ieee_fraud_bronze/mp5_landing/ba
tch_000000/_started_8299349501481159316
_started_8299349501481159316 0 1792613662000
dbfs:/Volumes/workspace/default/ieee_fraud_bronze/mp5_landing/ba
tch_000000/part-00000-tid-8299349501481159316-32792948-4c00-
4b24-b69a-ec147a162d8c-165-1.c000.snappy.parquet part-00000-tid-
8299349501481159316-32792948-4c00-4b24-b69a-ec147a162d8c-
165-1.c000.snappy.parquet 81214 1792613663000////////////////Rows
in first batch: 10
root
|-- TransactionID: long (nullable = true)
|-- isFraud: integer (nullable = true)
|-- TransactionDT: long (nullable = true)
|-- TransactionAmt: double (nullable = true)
|-- ProductCD: string (nullable = true)
|-- card1: string (nullable = true)
|-- card2: string (nullable = true)
|-- card3: string (nullable = true)
|-- card4: string (nullable = true)
|-- card5: string (nullable = true)
|-- card6: string (nullable = true)
|-- addr1: string (nullable = true)
|-- addr2: string (nullable = true)
|-- dist1: string (nullable = true)
|-- dist2: string (nullable = true)
|-- P_emaildomain: string (nullable = true)

```
|-- R_emaildomain: string (nullable = true)
|-- C1: string (nullable = true)
|-- C2: string (nullable = true)
|-- C3: string (nullable = true)
|-- C4: string (nullable = true)
|-- C5: string (nullable = true)
|-- C6: string (nullable = true)
|-- C7: string (nullable = true)
|-- C8: string (nullable = true)
|-- C9: string (nullable = true)
|-- C10: string (nullable = true)
|-- C11: string (nullable = true)
|-- C12: string (nullable = true)
|-- C13: string (nullable = true)
|-- C14: string (nullable = true)
|-- D1: string (nullable = true)
|-- D2: string (nullable = true)
|-- D3: string (nullable = true)
|-- D4: string (nullable = true)
|-- D5: string (nullable = true)
|-- D6: string (nullable = true)
|-- D7: string (nullable = true)
|-- D8: string (nullable = true)
|-- D9: string (nullable = true)
|-- D10: string (nullable = true)
|-- D11: string (nullable = true)
|-- D12: string (nullable = true)
|-- D13: string (nullable = true)
|-- D14: string (nullable = true)
|-- D15: string (nullable = true)
|-- M1: string (nullable = true)
|-- M2: string (nullable = true)
|-- M3: string (nullable = true)
|-- M4: string (nullable = true)
|-- M5: string (nullable = true)
|-- M6: string (nullable = true)
|-- M7: string (nullable = true)
|-- M8: string (nullable = true)
|-- M9: string (nullable = true)
|-- V1: string (nullable = true)
|-- V2: string (nullable = true)
|-- V3: string (nullable = true)
```

```
|-- V4: string (nullable = true)
|-- V5: string (nullable = true)
|-- V6: string (nullable = true)
|-- V7: string (nullable = true)
|-- V8: string (nullable = true)
|-- V9: string (nullable = true)
|-- V10: string (nullable = true)
|-- V11: string (nullable = true)
|-- V12: string (nullable = true)
|-- V13: string (nullable = true)
|-- V14: string (nullable = true)
|-- V15: string (nullable = true)
|-- V16: string (nullable = true)
|-- V17: string (nullable = true)
|-- V18: string (nullable = true)
|-- V19: string (nullable = true)
|-- V20: string (nullable = true)
|-- V21: string (nullable = true)
|-- V22: string (nullable = true)
|-- V23: string (nullable = true)
|-- V24: string (nullable = true)
|-- V25: string (nullable = true)
|-- V26: string (nullable = true)
|-- V27: string (nullable = true)
|-- V28: string (nullable = true)
|-- V29: string (nullable = true)
|-- V30: string (nullable = true)
|-- V31: string (nullable = true)
|-- V32: string (nullable = true)
|-- V33: string (nullable = true)
|-- V34: string (nullable = true)
.....
|-- event_ts: timestamp (nullable = true)
```

ChatGPT said:

You said:

```
Active: False
Exception: None
Recent progress: [{
  "id": "c26b9b30-2930-4c7f-afa5-6e2facedd467",
```

```
  "runId": "9b9e72b7-031f-40be-a623-1948d686b80f",
  "name": null,
  "timestamp": "2026-10-21T21:01:11.256Z",
  "batchId": 0,
  "batchDuration": 1092,
  "durationMs": {
    "triggerExecution": 1091,
    "queryPlanning": 93,
    "collectSourceMetrics": 0,
    "getBatch": 216,
    "commitOffsets": 98,
    "addBatch": 490,
    "latestOffset": 107,
    "walCommit": 166
  },
  "eventTime": {},
  "stateOperators": [],
  "sources": [
    {
      "description":
"FileStreamSource[dbfs:/Volumes/workspace/default/ieee_fraud_bronz
e/mp5_landing/batch_000000]",
      "startOffset": null,
      "endOffset": "{\"logOffset\":0}",
      "latestOffset": null,
      "numInputRows": 10,
      "inputRowsPerSecond": 0.0,
      "processedRowsPerSecond": 9.157509157509157,
      "metrics": {}
    }
  ],
  "sink": {
    "description":
"org.apache.spark.sql.execution.streaming.ConsoleTable$@3835ed34",
    "numOutputRows": 10,
    "metrics": {}
  },
  "observedMetrics": {},
  "rtmMetrics": null
}]
```

ChatGPT said:

You said:

Batch ID: 0 | Input rows: 40 | Output rows: 40

ChatGPT said:

You said:

mportant: make sure the `transactions_stream` used to create these window transformations includes `.option("recursiveFileLookup", "true")`. // Where do I change this

ChatGPT said:

You said:

after running this: OUTPUT_DIR_FINAL = f"
{BASE_DIR}/mp5_outputs_final"
CHECKPOINT_DIR_FINAL = f"{BASE_DIR}/mp5_checkpoints_final"

```
tumbling_query = (
    tumbling_60s.writeStream
    .format("parquet")
    .outputMode("append")
    .option("path", f"{OUTPUT_DIR_FINAL}/tumbling_60s")
    .option("checkpointLocation", f"
{CHECKPOINT_DIR_FINAL}/tumbling_60s")
    .trigger(availableNow=True)
    .start()
)

sliding_query = (
    sliding_5m_1m.writeStream
    .format("parquet")
    .outputMode("append")
    .option("path", f"{OUTPUT_DIR_FINAL}/sliding_5m_1m")
    .option("checkpointLocation", f"
{CHECKPOINT_DIR_FINAL}/sliding_5m_1m")
```

```
    .trigger(availableNow=True)
    .start()
)

tumbling_query.awaitTermination()
sliding_query.awaitTermination()//then ?
```

ChatGPT said:

You said:

```
Tumbling window rows: 1
Sliding window rows: 1
Tumbling results:
+----------------------------------------+----------------+------------
+-----------+
|window
|transaction_count|total_amount|fraud_count|
+----------------------------------------+----------------+------------
+-----------+
|{2025-01-01 00:00:00, 2025-01-01 00:01:00}|10              |1019.0      |0
|
+----------------------------------------+----------------+------------
+-----------+

Sliding results:
+----------------------------------------+----------------+------------
+-----------+
|window
|transaction_count|total_amount|fraud_count|
+----------------------------------------+----------------+------------
+-----------+
|{2024-12-31 23:56:00, 2025-01-01 00:01:00}|10              |1019.0      |0
|
+----------------------------------------+----------------+------------
+-----------+
```

ChatGPT said:

You said:

--- Tumbling ---
Batch: 0
Input rows: 40
Event time: {'min': '2025-01-01T00:00:00.000Z', 'avg': '2025-01-01T00:01:30.000Z', 'watermark': '1970-01-01T00:00:00.000Z', 'max': '2025-01-01T00:03:00.000Z'}
State operators: [{
"operatorName": "stateStoreSave",
"numRowsTotal": 4,
"numRowsUpdated": 4,
"allUpdatesTimeMs": 184,
"numRowsRemoved": 0,
"allRemovalsTimeMs": 1774,
"commitTimeMs": 9574,
"memoryUsedBytes": 263061619,
"numRowsDroppedByWatermark": 0,
"numShufflePartitions": 200,
"numStateStoreInstances": 200,
"customMetrics": {
"rocksdbNumSnapshotsAutoRepaired": 0,
"rocksdbTotalBytesWrittenByFlush": 0,
"rocksdbGetLatency": 0,
"rocksdbNumInternalColFamiliesKeys": 0,
"SnapshotLastUploaded.partition_157_default": -1,
"rocksdbCommitCompactLatency": 0,
"SnapshotLastUploaded.partition_33_default": -1,
"rocksdbTotalBytesWritten": 356,
"SnapshotLastUploaded.partition_172_default": -1,
"rocksdbPutLatency": 0,
"rocksdbForceSnapshotCount": 0,
"rocksdbNumInternalColumnFamilies": 0,
"rocksdbNumLoadedFromDfs": 0,
"rocksdbReadBlockCacheMissCount": 0,
"rocksdbPinnedBlocksMemoryUsage": 263020496,
"rocksdbCommitCheckpointLatency": 0,
"SnapshotLastUploaded.partition_139_default": -1,
"rocksdbZipFileBytesUncompressed": 0,
"rocksdbCommitFlushLatency": 0,
"rocksdbPutCount": 4,
"rocksdbCommitFileSyncLatencyMs": 8978,

```
      "rocksdbReadBlockCacheHitCount": 0,
      "rocksdbLoadLatencyMs": 3675,
      "rocksdbNumReplayChangelogFiles": 0,
      "rocksdbTotalBytesReadByCompaction": 0,
      "rocksdbTotalBytesWrittenByCompaction": 0,
      "rocksdbSaveZipFilesLatencyMs": 0,
      "rocksdbBytesCopied": 0,
      "rocksdbWriterStallLatencyMs": 0,
      "rocksdbTotalBytesRead": 0,
      "rocksdbChangeLogWriterCommitLatencyMs": 8974,
      "rocksdbGetCount": 8,
      "SnapshotLastUploaded.partition_178_default": -1,
      "rocksdbFilesReused": 0,
      "rocksdbTotalCompactionLatencyMs": 0,
      "rocksdbLoadFromSnapshotLatencyMs": 0,
      "rocksdbFilesCopied": 0,
      "rocksdbSstFileSize": 0,
      "rocksdbReplayChangeLogLatencyMs": 0,
      "rocksdbTotalBytesReadThroughIterator": 296,
      "rocksdbNumExternalColumnFamilies": 0,
      "numRowsReadDuringEviction": 4,
      "rocksdbTotalFlushLatencyMs": 0
    },
    "snapshotCustomMetricNames": [
      "SnapshotLastUploaded.partition_157_default",
      "SnapshotLastUploaded.partition_172_default",
      "SnapshotLastUploaded.partition_178_default",
      "SnapshotLastUploaded.partition_33_default",
      "SnapshotLastUploaded.partition_139_default",
      "SnapshotLastUploaded.partition_64_default",
      "SnapshotLastUploaded.partition_28_default",
      "SnapshotLastUploaded.partition_193_default",
      "SnapshotLastUploaded.partition_95_default",
      "SnapshotLastUploaded.partition_100_default",
      "SnapshotLastUploaded.partition_150_default",
      "SnapshotLastUploaded.partition_29_default",
      "SnapshotLastUploaded.partition_170_default",
      "SnapshotLastUploaded.partition_52_default",
      "SnapshotLastUploaded.partition_94_default",
      "SnapshotLastUploaded.partition_118_default",
      "SnapshotLastUploaded.partition_171_default",
      "SnapshotLastUploaded.partition_158_default",
```

"SnapshotLastUploaded.partition_13_default",
"SnapshotLastUploaded.partition_122_default",
"SnapshotLastUploaded.partition_125_default",
"SnapshotLastUploaded.partition_104_default",
"SnapshotLastUploaded.partition_30_default",
"SnapshotLastUploaded.partition_60_default",
"SnapshotLastUploaded.partition_177_default",
"rocksdbNumExternalColumnFamilies",
"rocksdbNumInternalColumnFamilies",
"SnapshotLastUploaded.partition_120_default",
"SnapshotLastUploaded.partition_192_default",
"SnapshotLastUploaded.partition_180_default",
"SnapshotLastUploaded.partition_162_default",
"SnapshotLastUploaded.partition_182_default",
"SnapshotLastUploaded.partition_145_default",
"SnapshotLastUploaded.partition_103_default",
"SnapshotLastUploaded.partition_46_default",
"SnapshotLastUploaded.partition_90_default",
"SnapshotLastUploaded.partition_191_default",
"SnapshotLastUploaded.partition_73_default",
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        "SnapshotLastUploaded.partition_57_default",
        "SnapshotLastUploaded.partition_167_default"
    ]
}]
Batch: 1
Input rows: 0
Event time: {'watermark': '2025-01-01T00:01:00.000Z'}
State operators: [{
    "operatorName": "stateStoreSave",
    "numRowsTotal": 7,
    "numRowsUpdated": 0,
    "allUpdatesTimeMs": 34,
    "numRowsRemoved": 1,
    "allRemovalsTimeMs": 906,
    "commitTimeMs": 9581,
    "memoryUsedBytes": 269076840,
    "numRowsDroppedByWatermark": 0,
    "numShufflePartitions": 200,
    "numStateStoreInstances": 200,
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        "rocksdbGetLatency": 0,
        "rocksdbNumInternalColFamiliesKeys": 0,
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        "rocksdbCommitCompactLatency": 0,
        "SnapshotLastUploaded.partition_33_default": -1,
        "rocksdbTotalBytesWritten": 55,
        "SnapshotLastUploaded.partition_172_default": -1,
        "rocksdbPutLatency": 0,
        "rocksdbForceSnapshotCount": 0,
        "rocksdbNumInternalColumnFamilies": 0,
        "rocksdbNumLoadedFromDfs": 0,
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```

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      "rocksdbTotalBytesRead": 33,
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      "SnapshotLastUploaded.partition_57_default",
      "SnapshotLastUploaded.partition_167_default"
    ]
}]
```


ChatGPT said:

You said:

Tumbling
batch = 2 | watermark = 2025-01-01T00:01:00.000Z | input = 40

Sliding
batch = 2 | watermark = 2025-01-01T00:01:00.000Z | input = 40

ChatGPT said:

You said:

: Generate the `readStream / withWatermark / groupBy(window(...))` skeleton with an explicit schema; audit it for schema inference, missing watermarks, unbounded state, and non-

idempotent sinks. I think we have done everything up to 'missing watermarks', 'unbounded state', 'non-idempotent sinks'

ChatGPT said:

You said:

--- Tumbling ---
Batch: 2
Watermark: 2025-01-01T00:01:00.000Z
State rows: 3
Updated: 3
Removed: 0
Dropped by watermark: 1

--- Sliding ---
Batch: 2
Watermark: 2025-01-01T00:01:00.000Z
State rows: 7
Updated: 7
Removed: 0
Dropped by watermark: 1

ChatGPT said:

You said:

is the main problem 'availableNow' = True ?

ChatGPT said:

You said:

File: part-00000-0b2da5b1-8ee7-4211-9b61-d5855e1b3f2a.c000.snappy.parquet -> Contains 0 rows File: part-00000-f8d4ac6b-d614-466a-857d-4efa30b6bb6e.c000.snappy.parquet -> Contains 0 rows File: part-00000-fd0b99a9-6227-4f1d-b24c-a56783026dd5.c000.snappy.parquet -> Contains 0 rows File: part-00082-f7987a46-7af1-4ee7-a4eb-18bdb85cd010.c000.snappy.parquet -> Contains 1 rows

svgsvg
// I thought we put in 4 rows. why is it only the last one that has ?

ChatGPT said:

You said:

okay, I get it now. Can you explain to me how we ended up with one row. Only 10 transactions fit into the window ?

ChatGPT said:

You said:

so when did withwatermark run, it started at 00:00 right ?

ChatGPT said:

You said:

so I guess we need a bigger watermark to preserve everything

ChatGPT said:

You said:

let's move on

ChatGPT said:

You said:

what about: unbounded state, and non-idempotent sinks

ChatGPT said:

You said:

what is unbounded state ?

ChatGPT said:

You said:

what is non-idempotent sink risk

ChatGPT said:

You said:

continue

ChatGPT said:

You said:

1. **Event-Time Calibration:** Replay the landing zone, measure the lateness distribution `ingest_ts - ts_end` (p50/p95/p99), and derive the watermark delay from it.

ChatGPT said:

You said:

Replay your records as rate-controlled micro-batches into a file-source landing directory,// Would that be part of this, or is it outside scope

ChatGPT said:

You said:

okay. Let's go

ChatGPT said:

You said:

no, let's rewrite event_ts: event_time = (
    F.to_timestamp(F.lit("2025-01-01 00:00:00"))
    + F.expr(f"INTERVAL {batch_index * 60} SECONDS")
)

ChatGPT said:

You said:

```
from pyspark.sql import functions as F

LANDING_DIR =
"/Volumes/workspace/default/ieee_fraud_bronze/mp5_landing"

# List the batch directories
display(dbutils.fs.ls(LANDING_DIR))

# Read the Parquet replay batches
landing_df = (
    spark.read
    .format("parquet")
    .option("recursiveFileLookup", "true")
    .load(LANDING_DIR)
)

# Check the schema and sample records
landing_df.printSchema()
display(landing_df.limit(10))

# Check row count and available timestamp fields
print("Total rows:", landing_df.count())
print("Timestamp fields:", [
    c for c in landing_df.columns
    if "ts" in c.lower() or "time" in c.lower()
])//returned: TransactionID  isFraud  TransactionDT  TransactionAmt
ProductCD  card1
2987010  0  86549  75.887  C  16496
```

```
2987011  0  86555  16.495 C  4461
2987012  0  86564  50     W 3786
2987013  0  86585  40     W 12866
2987014  0  86596  10.5   W 11839
2987015  0  86618  57.95    W 7055
2987016  0  86620  30     H  1790
2987017  0  86668  100    H  11492
2987018  0  86725  47.95    W 4663
2987019  0  86730  186    W 7005
```

ChatGPT said:

You said:

```
# 8. Pause before writing the next batch.
    if batch_index < MAX_NEW_BATCHES:
        time.sleep(DELAY_SECONDS)// yes it does
```

ChatGPT said:

You said:

can we not worry about having lateness increase or become negative, this is not real world data, and acting like that make this task more complicated

ChatGPT said:

You said:

```
import time
MAX_NEW_BATCHES = 3
BATCH_SIZE = 10
DELAY_SECONDS = 10
last_id = batch.agg(F.max("TransactionID")).first()[0]
SOURCE_TABLE = "workspace.default.ieee_train_transaction"
source = spark.table(SOURCE_TABLE)
from datetime import datetime

replay_start_time = datetime.now().strftime("%Y-%m-%d %H:%M:%S")
```

```
for batch_index in range(MAX_NEW_BATCHES + 1):

    # 1. Select the next 10 transactions.
    next_batch = (
        source
        .filter(F.col("TransactionID") > last_id)
        .orderBy("TransactionID")
        .limit(BATCH_SIZE)
    )

    # 2. Stop if there are no more records.
    if next_batch.count() == 0:
        print("No more transactions to replay.")
        break

    # 3. Give this batch a synthetic event timestamp.
    # Each batch starts one event-time minute after the previous one.
    event_time = (F.to_timestamp(F.lit(replay_start_time))
    + F.expr(f"INTERVAL {batch_index * 60} SECONDS"))


    ingest_time = datetime.now().replace(microsecond=0)
    next_batch = next_batch.withColumn("event_ts",
event_time).withColumn(
        "ts_end",
        F.col("event_ts")
    ).withColumn(
        "ingest_ts",
        F.to_timestamp(F.lit(ingest_time.strftime("%Y-%m-%d
%H:%M:%S")))
    )

    # 4. Build a unique directory for this batch.
    batch_name = f"batch_{batch_index:06d}"
    batch_path = f"{INPUT_DIR}/{batch_name}"

    # 5. Refuse to overwrite an existing batch.
    existing_paths = {
        item.path.rstrip("/")
        for item in dbutils.fs.ls(INPUT_DIR)
    }
```

```
    # if batch_path.rstrip("/") in existing_paths:
    #     raise RuntimeError(
    #         f"{batch_path} already exists; refusing to overwrite it."
    #     )

    # 6. Write the records as Parquet.
    # next_batch.write.format("parquet").save(batch_path)

next_batch.write.format("parquet").mode("overwrite").save(batch_path
)



    # 7. Advance the cursor so the next iteration selects new records.
    last_id = next_batch.agg(
        F.max("TransactionID")
    ).first()[0]

    print(
        f"Wrote {batch_name}: "
        f"{next_batch.count()} rows, "
        f"last TransactionID = {last_id}"
    )

    # 8. Pause before writing the next batch.
    if batch_index < MAX_NEW_BATCHES:
        time.sleep(DELAY_SECONDS)

print("Replay finished.")// here i my updated, any changes I need to
make ?
```

ChatGPT said:

You said:

where was `F` defined again ?

ChatGPT said:

You said:

1. **Event-Time Calibration:** Replay the landing zone, measure the lateness distribution `ingest_ts - ts_end` (p50/p95/p99), and derive the watermark delay from it.// Let's do this part

ChatGPT said:

You said:

```
+------------+-----------+-----------+-----------+----------------------
+
|record_count|p50_seconds|p95_seconds|p99_seconds|negative_latene
ss_count|
+------------+-----------+-----------+-----------+----------------------
+
|40          |-83        |1          |1          |30                    |
+------------+-----------+-----------+-----------+----------------------
+
```

ChatGPT said:

You said:

1. **Fault Injection:** Set `checkpointLocation`, write to an append-mode Parquet sink, kill the query mid-run, restart it, and reconcile row counts and window keys.

ChatGPT said:

You said:

Query started: d59ab84d-d592-477e-9c6a-07ebc842554c
Status: {'message': 'Initializing sources', 'isDataAvailable': False, 'isTriggerActive': False}

ChatGPT said:

You said:

the cell finished running

ChatGPT said:

You said:

Active: False
Status: {'message': 'Stopped', 'isDataAvailable': False, 'isTriggerActive': False}
Recent progress: [{

```
  "id": "d59ab84d-d592-477e-9c6a-07ebc842554c",
  "runId": "a8169d15-79ed-408c-a6c9-05d0917624dd",
  "name": null,
  "timestamp": "2026-10-22T03:36:04.475Z",
  "batchId": 0,
  "batchDuration": 10786,
  "durationMs": {
    "triggerExecution": 10785,
    "queryPlanning": 648,
    "collectSourceMetrics": 0,
    "getBatch": 149,
    "commitOffsets": 107,
    "addBatch": 9668,
    "latestOffset": 107,
    "walCommit": 200
  },
  "eventTime": {
    "min": "2026-10-22T03:09:41.000Z",
    "avg": "2026-10-22T03:09:41.000Z",
    "watermark": "1970-01-01T00:00:00.000Z",
    "max": "2026-10-22T03:09:41.000Z"
  },
  "stateOperators": [
    {
      "operatorName": "stateStoreSave",
      "numRowsTotal": 1,
      "numRowsUpdated": 1,
      "allUpdatesTimeMs": 90,
      "numRowsRemoved": 0,
      "allRemovalsTimeMs": 1734,
```

```
"commitTimeMs": 8507,
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"numStateStoreInstances": 200,
"customMetrics": {
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... [*** WARNING: max output size exceeded, skipping output. ***] ...

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e/mp5_landing]",
          "startOffset": "{\"logOffset\":2}",
          "endOffset": "{\"logOffset\":3}",
          "latestOffset": null,
          "numInputRows": 10,
          "inputRowsPerSecond": 2.5176233635448138,
          "processedRowsPerSecond": 2.5056376847907793,
          "metrics": {}
        }
      ],
      "sink": {
        "description":
"FileSink[/Volumes/workspace/default/ieee_fraud_bronze/mp5_fault_t
est/outputs/tumbling_60s]",
        "numOutputRows": -1,
        "metrics": {}
      },
      "observedMetrics": {},
      "rtmMetrics": null
}, {
      "id": "d59ab84d-d592-477e-9c6a-07ebc842554c",
      "runId": "a8169d15-79ed-408c-a6c9-05d0917624dd",
      "name": null,
      "timestamp": "2026-10-22T03:36:27.475Z",
      "batchId": 4,
      "batchDuration": 3569,
      "durationMs": {
        "triggerExecution": 3569,
        "queryPlanning": 66,
        "collectSourceMetrics": 0,
        "getBatch": 1,
```

```
    "commitOffsets": 95,
    "addBatch": 3308,
    "latestOffset": 0,
    "walCommit": 170
  },
  "eventTime": {
    "watermark": "2026-10-22T03:11:41.000Z"
  },
  "stateOperators": [
    {
      "operatorName": "stateStoreSave",
      "numRowsTotal": 2,
      "numRowsUpdated": 0,
      "allUpdatesTimeMs": 27,
      "numRowsRemoved": 1,
      "allRemovalsTimeMs": 912,
      "commitTimeMs": 8831,
      "memoryUsedBytes": 269077844,
      "numRowsDroppedByWatermark": 0,
      "numShufflePartitions": 200,
      "numStateStoreInstances": 200,
      "customMetrics": {
        "rocksdbNumSnapshotsAutoRepaired": 0,
        "rocksdbTotalBytesWrittenByFlush": 0,
        "rocksdbGetLatency": 0,
        "rocksdbNumInternalColFamiliesKeys": 0,
        "SnapshotLastUploaded.partition_157_default": -1,
        "rocksdbCommitCompactLatency": 0,
        "SnapshotLastUploaded.partition_33_default": -1,
        "rocksdbTotalBytesWritten": 55,
        "SnapshotLastUploaded.partition_172_default": -1,
        "rocksdbPutLatency": 0,
        "rocksdbForceSnapshotCount": 0,
        "rocksdbNumInternalColumnFamilies": 0,
        "rocksdbNumLoadedFromDfs": 0,
        "rocksdbReadBlockCacheMissCount": 0,
        "rocksdbPinnedBlocksMemoryUsage": 269049000,
        "rocksdbCommitCheckpointLatency": 0,
        "SnapshotLastUploaded.partition_139_default": -1,
        "rocksdbZipFileBytesUncompressed": 0,
        "rocksdbCommitFlushLatency": 0,
        "rocksdbPutCount": 1,
```

```
      "rocksdbCommitFileSyncLatencyMs": 8275,
      "rocksdbReadBlockCacheHitCount": 0,
      "rocksdbLoadLatencyMs": 3538,
      "rocksdbNumReplayChangelogFiles": 0,
      "rocksdbTotalBytesReadByCompaction": 0,
      "rocksdbTotalBytesWrittenByCompaction": 0,
      "rocksdbSaveZipFilesLatencyMs": 0,
      "rocksdbBytesCopied": 0,
      "rocksdbWriterStallLatencyMs": 0,
      "rocksdbTotalBytesRead": 33,
      "rocksdbChangeLogWriterCommitLatencyMs": 8272,
      "rocksdbGetCount": 1,
      "SnapshotLastUploaded.partition_178_default": -1,
      "rocksdbFilesReused": 0,
      "rocksdbTotalCompactionLatencyMs": 0,
      "rocksdbLoadFromSnapshotLatencyMs": 0,
      "rocksdbFilesCopied": 0,
      "rocksdbSstFileSize": 0,
      "rocksdbReplayChangeLogLatencyMs": 0,
      "rocksdbTotalBytesReadThroughIterator": 222,
      "rocksdbNumExternalColumnFamilies": 0,
      "numRowsReadDuringEviction": 3,
      "rocksdbTotalFlushLatencyMs": 0
    },
    "snapshotCustomMetricNames": [
      "SnapshotLastUploaded.partition_157_default",
      "SnapshotLastUploaded.partition_172_default",
      "SnapshotLastUploaded.partition_178_default",
      "SnapshotLastUploaded.partition_33_default",
      "SnapshotLastUploaded.partition_139_default",
      "SnapshotLastUploaded.partition_64_default",
      "SnapshotLastUploaded.partition_28_default",
      "SnapshotLastUploaded.partition_193_default",
      "SnapshotLastUploaded.partition_95_default",
      "SnapshotLastUploaded.partition_100_default",
      "SnapshotLastUploaded.partition_150_default",
      "SnapshotLastUploaded.partition_29_default",
      "SnapshotLastUploaded.partition_170_default",
      "SnapshotLastUploaded.partition_52_default",
      "SnapshotLastUploaded.partition_94_default",
      "SnapshotLastUploaded.partition_118_default",
      "SnapshotLastUploaded.partition_171_default",
```

"SnapshotLastUploaded.partition_158_default",
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"SnapshotLastUploaded.partition_162_default",
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"SnapshotLastUploaded.partition_145_default",
"SnapshotLastUploaded.partition_103_default",
"SnapshotLastUploaded.partition_46_default",
"SnapshotLastUploaded.partition_90_default",
"SnapshotLastUploaded.partition_191_default",
"SnapshotLastUploaded.partition_73_default",
"SnapshotLastUploaded.partition_134_default",
"SnapshotLastUploaded.partition_153_default",
"SnapshotLastUploaded.partition_165_default",
"SnapshotLastUploaded.partition_79_default",
"SnapshotLastUploaded.partition_2_default",
"SnapshotLastUploaded.partition_78_default",
"SnapshotLastUploaded.partition_123_default",
"SnapshotLastUploaded.partition_199_default",
"SnapshotLastUploaded.partition_81_default",
"SnapshotLastUploaded.partition_40_default",
"SnapshotLastUploaded.partition_72_default",
"SnapshotLastUploaded.partition_27_default",
"SnapshotLastUploaded.partition_138_default",
"SnapshotLastUploaded.partition_92_default",
"SnapshotLastUploaded.partition_25_default",
"SnapshotLastUploaded.partition_148_default",
"SnapshotLastUploaded.partition_131_default",
"SnapshotLastUploaded.partition_126_default",
"SnapshotLastUploaded.partition_169_default",
"SnapshotLastUploaded.partition_84_default",
"SnapshotLastUploaded.partition_187_default",

"SnapshotLastUploaded.partition_188_default",
"SnapshotLastUploaded.partition_37_default",
"SnapshotLastUploaded.partition_68_default",
"SnapshotLastUploaded.partition_156_default",
"SnapshotLastUploaded.partition_181_default",
"SnapshotLastUploaded.partition_115_default",
"SnapshotLastUploaded.partition_76_default",
"SnapshotLastUploaded.partition_0_default",
"SnapshotLastUploaded.partition_1_default",
"SnapshotLastUploaded.partition_83_default",
"SnapshotLastUploaded.partition_154_default",
"SnapshotLastUploaded.partition_112_default",
"SnapshotLastUploaded.partition_174_default",
"SnapshotLastUploaded.partition_189_default",
"SnapshotLastUploaded.partition_11_default",
"SnapshotLastUploaded.partition_166_default",
"SnapshotLastUploaded.partition_159_default",
"SnapshotLastUploaded.partition_119_default",
"SnapshotLastUploaded.partition_106_default",
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"SnapshotLastUploaded.partition_32_default",
"SnapshotLastUploaded.partition_49_default",
"SnapshotLastUploaded.partition_55_default",
"SnapshotLastUploaded.partition_51_default",
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"SnapshotLastUploaded.partition_65_default",
"SnapshotLastUploaded.partition_14_default",
"SnapshotLastUploaded.partition_47_default",
"SnapshotLastUploaded.partition_142_default",
"SnapshotLastUploaded.partition_22_default",
"SnapshotLastUploaded.partition_133_default",
"SnapshotLastUploaded.partition_130_default",
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"SnapshotLastUploaded.partition_183_default",
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"SnapshotLastUploaded.partition_121_default",
"SnapshotLastUploaded.partition_26_default",
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"SnapshotLastUploaded.partition_42_default",
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"SnapshotLastUploaded.partition_109_default",
"SnapshotLastUploaded.partition_176_default",
"SnapshotLastUploaded.partition_160_default",
"SnapshotLastUploaded.partition_67_default",
"SnapshotLastUploaded.partition_161_default",
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"SnapshotLastUploaded.partition_91_default",
"SnapshotLastUploaded.partition_50_default",
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"SnapshotLastUploaded.partition_62_default",
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"SnapshotLastUploaded.partition_36_default",
"SnapshotLastUploaded.partition_31_default",
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"SnapshotLastUploaded.partition_89_default",
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"SnapshotLastUploaded.partition_124_default",
"SnapshotLastUploaded.partition_6_default",
"SnapshotLastUploaded.partition_15_default",
"SnapshotLastUploaded.partition_88_default",
"SnapshotLastUploaded.partition_132_default",

"SnapshotLastUploaded.partition_71_default",
"SnapshotLastUploaded.partition_20_default",
"SnapshotLastUploaded.partition_135_default",
"SnapshotLastUploaded.partition_141_default",
"SnapshotLastUploaded.partition_197_default",
"SnapshotLastUploaded.partition_87_default",
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"SnapshotLastUploaded.partition_56_default",
"SnapshotLastUploaded.partition_113_default",
"SnapshotLastUploaded.partition_107_default",
"SnapshotLastUploaded.partition_186_default",
"SnapshotLastUploaded.partition_8_default",
"SnapshotLastUploaded.partition_147_default",
"rocksdbNumInternalColFamiliesKeys",
"SnapshotLastUploaded.partition_75_default",
"SnapshotLastUploaded.partition_45_default",
"SnapshotLastUploaded.partition_10_default",
"SnapshotLastUploaded.partition_149_default",
"SnapshotLastUploaded.partition_108_default",
"SnapshotLastUploaded.partition_69_default",
"SnapshotLastUploaded.partition_80_default",
"SnapshotLastUploaded.partition_19_default",
"SnapshotLastUploaded.partition_77_default",
"SnapshotLastUploaded.partition_24_default",
"SnapshotLastUploaded.partition_59_default",
"SnapshotLastUploaded.partition_85_default",
"SnapshotLastUploaded.partition_96_default",
"SnapshotLastUploaded.partition_190_default",
"SnapshotLastUploaded.partition_38_default",
"SnapshotLastUploaded.partition_128_default",
"SnapshotLastUploaded.partition_114_default",
"SnapshotLastUploaded.partition_151_default",
"SnapshotLastUploaded.partition_179_default",
"SnapshotLastUploaded.partition_34_default",
"SnapshotLastUploaded.partition_164_default",
"SnapshotLastUploaded.partition_117_default",
"SnapshotLastUploaded.partition_196_default",
"SnapshotLastUploaded.partition_105_default",
"SnapshotLastUploaded.partition_93_default",
"SnapshotLastUploaded.partition_18_default",
"SnapshotLastUploaded.partition_194_default",
"SnapshotLastUploaded.partition_43_default",

```
            "SnapshotLastUploaded.partition_16_default",
            "SnapshotLastUploaded.partition_195_default",
            "SnapshotLastUploaded.partition_101_default",
            "SnapshotLastUploaded.partition_137_default",
            "SnapshotLastUploaded.partition_144_default",
            "SnapshotLastUploaded.partition_3_default",
            "SnapshotLastUploaded.partition_21_default",
            "SnapshotLastUploaded.partition_66_default",
            "SnapshotLastUploaded.partition_146_default",
            "rocksdbPinnedBlocksMemoryUsage",
            "SnapshotLastUploaded.partition_185_default",
            "SnapshotLastUploaded.partition_44_default",
            "SnapshotLastUploaded.partition_12_default",
            "SnapshotLastUploaded.partition_136_default",
            "SnapshotLastUploaded.partition_163_default",
            "rocksdbSstFileSize",
            "SnapshotLastUploaded.partition_61_default",
            "SnapshotLastUploaded.partition_155_default",
            "SnapshotLastUploaded.partition_57_default",
            "SnapshotLastUploaded.partition_167_default"
          ]
        }
    ],
    "sources": [
        {
            "description":
"FileStreamSource[dbfs:/Volumes/workspace/default/ieee_fraud_bronz
e/mp5_landing]",
            "startOffset": "{\"logOffset\":3}",
            "endOffset": "{\"logOffset\":3}",
            "latestOffset": null,
            "numInputRows": 0,
            "inputRowsPerSecond": 0.0,
            "processedRowsPerSecond": 0.0,
            "metrics": {}
        }
    ],
    "sink": {
        "description":
"FileSink[/Volumes/workspace/default/ieee_fraud_bronze/mp5_fault_t
est/outputs/tumbling_60s]",
        "numOutputRows": -1,
```

"metrics": {}
},
"observedMetrics": {},
"rtmMetrics": null
}]
Exception: None

ChatGPT said:

You said:

let's clear the directory, the run took 28 seconds, so if we interrupt after 18 seconds we should be fine

ChatGPT said:

You said:

I can't run two cells in a notebook at once

ChatGPT said:

You said:

it worked

ChatGPT said:

You said:

}]
Stop requested.
Active after interruption: False
Exception: None

ChatGPT said:

You said:

```
Output rows: 2
Distinct window keys: 2
Duplicate window keys:
+-----+---+-----+
|start|end|count|
+-----+---+-----+
+-----+---+-----+

Output window aggregates:
+------------------------------------------+-----------------+-------------
----+-----------+
|window                                    |transaction_count|total_amount
|fraud_count|
+------------------------------------------+-----------------+-------------
----+-----------+
|{2026-10-22 03:09:00, 2026-10-22 03:10:00}|10               |1019.0
|0          |
|{2026-10-22 03:10:00, 2026-10-22 03:11:00}|10
|614.7819999999999|0          |
+------------------------------------------+-----------------+-------------
----+-----------+
```

ChatGPT said:

You said:

```
Input transaction rows: 40
Expected window aggregates:
+------------------------------------------+-----------------+-------------
-----+-----------+
|window                                    |transaction_count|total_amount
|fraud_count|
+------------------------------------------+-----------------+-------------
-----+-----------+
|{2026-10-22 03:09:00, 2026-10-22 03:10:00}|10               |1019.0
|0          |
|{2026-10-22 03:10:00, 2026-10-22 03:11:00}|10
|614.7819999999999 |0          |
|{2026-10-22 03:11:00, 2026-10-22 03:12:00}|10               |827.74
|0          |
```

```
|{2026-10-22 03:12:00, 2026-10-22 03:13:00}|10
|1018.6900000000002|0          |
+-----------------------------------------+----------------+-------------
-----+-----------+

Streaming output aggregates:
+-----------------------------------------+----------------+-------------
----+-----------+
|window                                   |transaction_count|total_amount
|fraud_count|
+-----------------------------------------+----------------+-------------
----+-----------+
|{2026-10-22 03:09:00, 2026-10-22 03:10:00}|10              |1019.0
|0          |
|{2026-10-22 03:10:00, 2026-10-22 03:11:00}|10
|614.7819999999999|0          |
+-----------------------------------------+----------------+-------------
----+-----------+
```

ChatGPT said:

You said:

```
Streaming output rows: 2
Distinct window keys: 2
+-----------------------------------------+----------------+-------------
----+-----------+
|window                                   |transaction_count|total_amount
|fraud_count|
+-----------------------------------------+----------------+-------------
----+-----------+
|{2026-10-22 03:09:00, 2026-10-22 03:10:00}|10              |1019.0
|0          |
|{2026-10-22 03:10:00, 2026-10-22 03:11:00}|10
|614.7819999999999|0          |
+-----------------------------------------+----------------+-------------
----+-----------+
```

ChatGPT said:

You said:

Let's just stop here, and write the report

ChatGPT said:

You said:

- **Schema and Contracts:** Explicit `StructType` (`TranscriptChunk` for ASR; `CorpusChunk` for VDB: `record_id` , `track` , `source_id` , `ts_start` , `ts_end` , `text` , `ingest_ts` , plus `license_class` for corpus records), nullability justification, and the `deal` contracts.
- **Watermark and State Analysis:** Lateness percentiles, the chosen watermark delay and its completeness-vs-latency trade-off, the required output mode, and the state-growth bound.
- **Query Design:** 60-second tumbling and 5-minute/1-minute sliding windows computing record count, `approx_count_distinct` of tokens, top-10 terms, and mean ingest latency; sink design.
- **Fault-Tolerance Reconciliation:** Restart procedure; row-count and window-key reconciliation showing no duplicate or missing windows; which exactly-once guarantees come from checkpointing versus an idempotent sink.
- **Metrics and Reproducibility:** `StreamingQueryProgress` summary (`inputRowsPerSecond` , `processedRowsPerSecond` , `batchDuration` , `stateOperators.numRowsTotal`), pinned-environment statement, CI status, and the AssemblyAI quota log (ASR track). Throughput and latency targets are liveness properties (INV-5) assessed from the metrics log, not from contracts.
- **GenAI Audit:** Did you use the Generative AI Kit to synthesize this artifact? You must include your `GenAI.pdf` log or explicitly state "No GenAI Used."

ChatGPT said:

You said:

```
from datetime import datetime, timezone
from pyspark.sql import DataFrame
from pyspark.sql.types import (
    StructType,
    StructField,
    LongType,
    StringType,
    DoubleType,
    IntegerType,
    TimestampType,
)
from pyspark.sql import functions as F


EVENT_TIME_ANCHOR = datetime(2017, 12, 1, tzinfo=timezone.utc)
EVENT_TIME_ANCHOR_UNIX =
int(EVENT_TIME_ANCHOR.timestamp())


TRANSACTION_CHUNK_SCHEMA = StructType([
    # --- Identity / provenance ---
    StructField("record_id",   LongType(),    nullable=False),  #
TransactionID
    StructField("track",       StringType(),  nullable=False),  # constant
"STRUCTURED_REPLAY"
    StructField("source_id",   StringType(),  nullable=False),  # source_file
+ extraction_date
    StructField("split",       StringType(),  nullable=False),  # "train" | "test"

    # --- Event time (derived) ---
    StructField("transaction_dt", LongType(),    nullable=False),  # raw
offset, kept for audit
    StructField("ts_start",       TimestampType(), nullable=False),  #
derived event_ts
    StructField("ts_end",         TimestampType(), nullable=False),  # ==
ts_start (point event)

    # --- Payload fields ---
    StructField("amount",      DoubleType(),  nullable=True),   #
TransactionAmt — verify
```

```
    StructField("product_cd",  StringType(),  nullable=True),   # verify
against full table
    StructField("card1",       IntegerType(), nullable=True),
    StructField("card2",       DoubleType(),  nullable=True),   # observed
null in sample row 1

    # --- Label (train only) ---
    StructField("is_fraud",    IntegerType(), nullable=True),   # non-
null+0/1 on train; null on test

    # --- Replay / streaming bookkeeping ---
    StructField("ingest_ts",   TimestampType(), nullable=False),  #
stamped at replay read time
])

def validate_bounded_chunking(df: DataFrame) -> DataFrame:
    """
    Returns rows that VIOLATE the adapted BoundedChunking contract.
    An empty result means the contract holds for this batch.
    """
    chunk_duration = F.col("ts_end").cast("long") -
F.col("ts_start").cast("long")
    violations = df.withColumn("_chunk_duration_sec",
chunk_duration).filter(
        (F.col("ts_start") > F.col("ts_end"))  # ts_start <= ts_end required
        | (F.col("_chunk_duration_sec") != MAX_CHUNK_DURATION_SEC)
    )
    return violations


def validate_provenance(df: DataFrame, source_whitelist: set = None) -
> DataFrame:
    """
    Returns rows that fail the provenance gate: missing required
identity
    fields, or (if source_whitelist is given) sourced from a Bronze
    source_id / extraction batch that isn't on the admitted list. This is
    the analogue of the ASR track's source-whitelist (INV-10) — it gates
    on WHERE the record came from, not on its payload values.
    """
    missing = (
        F.col("record_id").isNull()
```

```
        | F.col("source_id").isNull()
        | F.col("ts_start").isNull()
    )
    if source_whitelist:
        missing = missing | (~F.col("source_id").isin(list(source_whitelist)))
    return df.filter(missing)


def validate_label(df: DataFrame) -> DataFrame:
    """
    Returns rows that violate the label contract:
      - split == 'train'  -> is_fraud must be non-null and in {0, 1}
      - split == 'test'   -> is_fraud must be null
    """
    bad_train = (F.col("split") == "train") & (
        F.col("is_fraud").isNull() | (~F.col("is_fraud").isin([0, 1]))
    )
    bad_test = (F.col("split") == "test") & F.col("is_fraud").isNotNull()
    return df.filter(bad_train | bad_test)


def validate_uniqueness(df: DataFrame):
    """
    Returns (duplicate_ids_df, null_id_count) for the provenance /
    duplicate-detection gate. record_id (TransactionID) must be unique
    and non-null within a split.
    """
    null_id_count = df.filter(F.col("record_id").isNull()).count()
    dupes = (
        df.groupBy("record_id")
        .count()
        .filter(F.col("count") > 1)
    )
    return dupes, null_id_count


# ---------------------------------------------------------------------------
# 3. Nullability verification harness (run against FULL Bronze tables)
# ---------------------------------------------------------------------------

def verify_nullability(df: DataFrame, columns=None) -> DataFrame:
    """
```

```
    Computes null counts and null fraction per column so that each
    nullable= choice in TRANSACTION_CHUNK_SCHEMA can be
justified with
    numbers rather than asserted from a 5-row sample.

    Run this against the FULL train/test Bronze tables before finalizing
    the schema above. Update nullability + whitelist/doc comments
based on
    the actual output.
    """
    cols = columns or df.columns
    total = df.count()
    agg_exprs = [
        F.sum(F.col(c).isNull().cast("int")).alias(c) for c in cols
    ]
    null_counts = df.agg(*agg_exprs).collect()[0].asDict()
    rows = [
        (c, null_counts[c], total, (null_counts[c] / total if total else None))
        for c in cols
    ]
    return rows  # list of (column, null_count, total_rows, null_fraction)


# ---------------------------------------------------------------------------
# 4. Convenience: run every contract check in one pass
# ---------------------------------------------------------------------------

def run_all_contracts(df: DataFrame) -> dict:
    """
    Runs all deal-contract checks against a (replayed) micro-batch and
    returns a dict of violation counts. Intended to be called from the
    streaming pipeline's foreachBatch or from a batch-mode audit in
    Step 1, before the scaffold in Step 2 is wired up.
    """
    dupes, null_ids = validate_uniqueness(df)
    return {
        "bounded_chunking_violations":
validate_bounded_chunking(df).count(),
        "provenance_violations": validate_provenance(df).count(),
        "label_violations": validate_label(df).count(),
        "duplicate_record_ids": dupes.count(),
        "null_record_ids": null_ids,
```

```
    }


# ---------------------------------------------------------------------------
# 5. Runnable nullability audit — Step 1, against the FULL Bronze tables
# ---------------------------------------------------------------------------
#
# Usage (from a Databricks notebook cell, or spark-submit with the table
# names adjusted):
#
#   python contracts.py
#
# This prints a null-count report per raw Bronze column for each of the
# four tables, which is the evidence you paste into report.md's "Schema
# and Contracts" section to justify each nullable= choice in
# TRANSACTION_CHUNK_SCHEMA — not the 5-row sample.

RAW_BRONZE_TABLES = {
    "train_transaction": "workspace.default.ieee_train_transaction",
    "test_transaction":  "workspace.default.ieee_test_transaction",
    "train_identity":    "workspace.default.ieee_train_identity",
    "test_identity":     "workspace.default.ieee_test_identity",
}

# Columns to check per table. Keep this list short and targeted at the
# fields TRANSACTION_CHUNK_SCHEMA actually depends on, rather than every
# V-feature column — those are out of scope for this contract.
COLUMNS_TO_CHECK = {
    "train_transaction": ["TransactionID", "isFraud", "TransactionDT", "TransactionAmt", "ProductCD", "card1", "card2"],
    "test_transaction":  ["TransactionID", "TransactionDT", "TransactionAmt", "ProductCD", "card1", "card2"],
    "train_identity":    ["TransactionID"],
    "test_identity":     ["TransactionID"],
}
```

```
def print_nullability_report(spark) -> None:
    """
    Runs verify_nullability() against each real Bronze table and prints a
    plain-text report. Run this once, capture the printed output, and
paste
    the numbers (not a screenshot) into report.md.
    """
    for label, table_name in RAW_BRONZE_TABLES.items():
        print(f"\n=== {label} ({table_name}) ===")
        df = spark.table(table_name)
        cols = COLUMNS_TO_CHECK[label]
        rows = verify_nullability(df, columns=cols)
        print(f"{'column':<18}{'null_count':>12}{'total_rows':>12}
{'null_fraction':>16}")
        for col, null_count, total, frac in rows:
            frac_str = f"{frac:.4%}" if frac is not None else "n/a"
            print(f"{col:<18}{null_count:>12}{total:>12}{frac_str:>16}")


if __name__ == "__main__":
    # Assumes a Databricks / Spark environment where spark is
already
    # available. If running outside Databricks, construct a SparkSession
    # first, e.g.:
    #
    #   from pyspark.sql import SparkSession
    #   spark = SparkSession.builder.getOrCreate()
    #
    print_nullability_report(spark)  # noqa: F821 — spark provided by
runtime// can you rewrite this for what we have just done
```

ChatGPT said:

You said:

Objective: Your submission must be a single `report.md` (at most 3 pages) delivered with Git tag `mp5-submission` (along with your code, `contracts.py`, `metrics.jsonl`, and reconciliation notebook) containing the following sections exactly as named below.

- **Schema and Contracts:** Explicit `StructType` (`TranscriptChunk` for ASR; `CorpusChunk` for VDB: `record_id` , `track` , `source_id` , `ts_start` , `ts_end` , `text` , `ingest_ts` , plus `license_class` for corpus records), nullability justification, and the `deal` contracts.
- **Watermark and State Analysis:** Lateness percentiles, the chosen watermark delay and its completeness-vs-latency trade-off, the required output mode, and the state-growth bound.
- **Query Design:** 60-second tumbling and 5-minute/1-minute sliding windows computing record count, `approx_count_distinct` of tokens, top-10 terms, and mean ingest latency; sink design.
- **Fault-Tolerance Reconciliation:** Restart procedure; row-count and window-key reconciliation showing no duplicate or missing windows; which exactly-once guarantees come from checkpointing versus an idempotent sink.
- **Metrics and Reproducibility:** `StreamingQueryProgress` summary (`inputRowsPerSecond` , `processedRowsPerSecond` , `batchDuration` , `stateOperators.numRowsTotal`), pinned-environment statement, CI status, and the AssemblyAI quota log (ASR track). Throughput and latency targets are liveness properties (INV-5) assessed from the metrics log, not from contracts.
- **GenAI Audit:** Did you use the Generative AI Kit to synthesize this artifact? You must include your `GenAI.pdf` log or explicitly state "No GenAI Used."

//Let's write the report

ChatGPT said:

You said:

can't I export this chaat

ChatGPT said:

You said:

can you specify the deal contracts with references

ChatGPT said:

You said:

by record_id do you mean transactionid

ChatGPT said:

You said:

what is source_id

ChatGPT said:

You said:

where did we define source_id, I cant see it in the notebook

ChatGPT said:

You said:

The sliding window overlaps five one-minute intervals per event, increasing aggregation work relative to a tumbling window.
// What does this mean ? what does the sliding window have to do with the tumbling window ?

ChatGPT said: